



SLAM-BASED AUV DEPLOYMENT

Presented in
SMART INDIAN HACKATHON 2024

By

-Ashutosh Shrivastav

-Hrituraj Singh

-Anindya Mukherjee.



IN SEARCH OF INNOVATION

BUILDNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNeo prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

Problem Statement - Validation of deployment of SLAM-enabled AUVs for Real-time Draft Monitoring and Siltation Pattern Prediction

AUTONOMOUS UNDERWATER VEHICLE (AUV)

To establish a comprehensive real-time monitoring framework for draft clearance in the berths and navigational channels of Cochin Port by leveraging SLAM-based Autonomous Underwater Vehicles (AUVs) for high-resolution bathymetric mapping and integrating ERA-5 reanalysis data for predictive modeling. This initiative aims to enhance the accuracy, frequency, and reliability of depth assessments, addressing limitations in traditional data acquisition methods and improving navigational safety and port operational efficiency.

METHODOLOGY:

1. Observation was made that past data acquisition methods were infrequent and sparsely distributed
2. Identification of SLAM-based AUVs as a solution.
3. ERA-5 integration for improved prediction model
4. Preliminary particulars identification

SPECIFICATIONS:

1. DOF: 6
2. Main Hull: Custom Aluminum Milling
3. Electronics Bay: Custom Aluminum Laser-cut
4. Maximum dive depth: 24m
5. LOA: 800mm
6. Beam: 350mm

FEATURES:

1. Omni-directional thrusters
2. IoT based communications syst
3. SLAM-based imagery
4. TOS integration

